

DHIRAJ YESHWANT NIMJE

Machine Learning / Data Engineering • Mumbai, India

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SUMMARY

MCA candidate (Somaiya Vidyavihar University, 2027) with a **B.Sc. Mathematics** foundation in statistics and linear algebra, building applied ML systems across retrieval-augmented generation and on-device computer vision — including a **neural network implemented from scratch** (backpropagation, Adam) that trains and runs entirely in the browser. Three end-to-end projects shipped with validated benchmarks and live deployments; **runner-up at KJSIM Hackathon 2026** and a Smart India Hackathon 2026 pre-qualifier.

SKILLS

Languages Python, SQL, TypeScript, Kotlin

ML & Data scikit-learn, XGBoost, pandas, NumPy

Neural Networks Deep learning — feedforward networks built from scratch: backpropagation, Adam, minibatch training, in-browser inference

GenAI / RAG Hugging Face Sentence Transformers, ChromaDB, Gemini API

Databases PostgreSQL, SQLAlchemy 2.0

Frameworks FastAPI, Next.js, React, Docker

Computer Vision OpenCV, MediaPipe Tasks Vision, CameraX, facial landmark metrics

EDUCATION

Master of Computer Applications (MCA), 2-Year — Computer Science — KJ Somaiya Institute of Management, Mumbai | Somaiya Vidyavihar University
CGPA 9.29 / 10 • 2027 (expected)

B.Sc. Mathematics — B.K. Birla College of Arts, Science & Commerce (Autonomous), Kalyan | MSBSHSE
73.13% • 2025

Higher Secondary (Class XII) — Shree Shantilal Karamshi Somaiya Vinay Mandir Junior College | MSBSHSE
52.17% • 2022

Secondary (Class X) — St Mary's High School | MSBSHSE
88.60% • 2020

PROJECTS

PrakritiAI — Ayurvedic Prakriti Classification & Research System

Python • scikit-learn • XGBoost • pandas • NumPy • FastAPI • TypeScript • React • OpenPyXL • Docker | github.com/jerry11-tech/prakritiai
| Live: jerry11-tech.github.io/prakritiai

- Reached **96.69% accuracy (macro-F1 0.9670)** on a frozen, participant-level 25% unseen test split held out of all hyperparameter tuning and model selection; Cohen's $\kappa = 0.9503$ and expert inter-rater reliability Fleiss' $\kappa = 0.7016$.
- Built **DoshaNet**, a feedforward neural network written **from scratch in TypeScript** — hand-derived backpropagation, Adam optimizer (lr 0.004, batch 128, 64-unit hidden layer) over a 4,000-sample generated corpus with a seeded 25% hold-out — whose trained weights ship as JSON for fully in-browser inference with no server or Python runtime.
- Benchmarked **6 algorithms** (Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting, XGBoost) under stratified 5-fold cross-validation, reporting model evaluation across accuracy, macro-F1 and Cohen's κ , with versioned model, preprocessing and feature-schema artifacts.
- Applied **feature engineering** to construct the predictor vector, deliberately excluding demographic and health variables to prevent leakage, and designed a **non-circular ML formulation** that predicts independent expert ground-truth labels rather than self-referential questionnaire rules.
- Built a **blind expert-consensus portal** with disagreement flagging and multi-rater reliability computation, backed by a 5-sheet Excel audit engine maintaining immutable change history and complete audit documentation, with a verify / re-verify workflow and automated unit, integration and RBAC security test suites.
- Shipped a **live deployment** with automated 3-tier disaster backup (SQLite snapshot → Excel mirror → JSON audit dump) and versioned model lineage.

LegalSathi AI — Multilingual Legal Guidance & Action Planner (RAG)

Python 3.12 • FastAPI • Hugging Face Sentence Transformers • ChromaDB • Gemini API • PostgreSQL • SQLAlchemy 2.0 • Next.js 14 • Docker
| github.com/jerry11-tech/legal-sathi-ai

- Built a **natural language processing (NLP) pipeline** for retrieval-augmented generation, answering Indian legal queries in **11+ languages**, grounding responses in a ChromaDB **vector database** indexed with Hugging Face all-MiniLM-L6-v2 Sentence Transformers embeddings, with Gemini handling grounded response generation.
- Extended beyond Q&A with **legal document generation** (FIRs, rental notices) and an ingestion **data pipeline**, backed by PostgreSQL persistence for user accounts and full query history.
- Deployed a **production stack on free-tier infrastructure** — Next.js/React on Vercel, FastAPI + Gunicorn on Render, PostgreSQL on Supabase/Neon — with JWT authentication, Argon2/bcrypt hashing and guest rate limiting.
- Built a **golden evaluation dataset and QA test plan** for RAG output quality — hand-labelled legal questions carrying expected statute, section and key-fact assertions, paired with a structured test matrix tracking priority and pass/fail status per module.

Blink Flow — Privacy-Preserving Gesture-Controlled Scrolling (Android)

Kotlin • CameraX • Google MediaPipe Tasks Vision • Android AccessibilityService | github.com/jerry11-tech/blink-flow

- Developed a touch-free accessibility tool that controls scrolling across any application — Reels, Shorts, PDFs, eBooks, web pages — via **double-blink, open-palm, fist and head-tilt gestures**.
- Implemented **real-time on-device vision** with MediaPipe FaceLandmarker and HandLandmarker, deriving gestures from Eye Aspect Ratio, hand openness and head-roll landmarks; frames are processed in local RAM and discarded immediately.
- Designed a **temporal state machine** (Idle → Detected → Confirmed → Cooldown) with 4-step per-user calibration and multi-frame validation to suppress false triggers and make gesture recognition reliable enough for daily use.
- Enforced a strict privacy posture: **zero storage, zero network calls, no accounts and no analytics SDKs**, with all processing and gesture dispatch performed locally through AccessibilityService.

HONORS & HACKATHONS

- Runner-Up — **KJSIM Hackathon 2026**, KJ Somaiya Institute of Management. Five-person team, end-to-end system ownership from data pipeline to deployed map UI.
- **Smart India Hackathon 2026 — Pre-qualifier** (Problem Statement 26162, Disaster Management).

CERTIFICATIONS

- Supervised Machine Learning: Classification — IBM via Coursera, 2026
- Supervised Machine Learning: Regression — IBM via Coursera, 2026
- Algorithms for Searching, Sorting and Indexing — University of Colorado Boulder via Coursera, 2025

TECHNICAL WRITING

- "Blink Flow: Privacy-Preserving Universal Gesture-Controlled Scrolling" — Medium, Aug 2026
- "LegalSathi AI: Architectural Design of a Multilingual Legal Guidance System" — Medium, Aug 2026
- "AI & Machine Learning Insights" — Medium, May 2026

INTERESTS

- AI and emerging technologies; AI-generated art • fitness, sports and travel